

# ACOUSTIC PROCESSING OF SPEECH

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## Acoustic processing:

Acoustic processing is commonly called feature extraction or signal analysis in the speech recognition literature.

## Feature Extraction/Signal Analysis:

The process of converting a speech signal into a vector of numbers representing a time slice. These vectors are called features.

### Types of Features:

- LPC features
- PLP features

The term 'features' refers to the vector of numbers that represent one time-slice of a speech signal. Several types of features are commonly used, including LPC features and PLP features.

## Spectral Features:

All of these are spectral features, which means that they represent the waveform in terms of the distribution of different frequencies that make up the waveform; such a distribution of frequencies is called a spectrum.

# What are Sound Waves?

- Sound waves are **changes in air pressure** caused by the vibration of air particles.

When we speak, air from our lungs passes through the vocal cords (glottis) and out through the mouth or nose — this movement **creates variations in air pressure.**

We can record or study these waves by plotting:

- **X-axis:** Time
- **Y-axis:** Change in air pressure
- So, a sound wave graph shows how air pressure changes over time.

# Waveform Representation

A waveform is a graph that measures the compression or rarefaction (uncompression) of air molecules.

- **Compression:** Air molecules are pushed together → higher air pressure
- **Rarefaction:** Air molecules are spread apart → lower air pressure

The wave on the graph moves **up and down** depending on these changes.

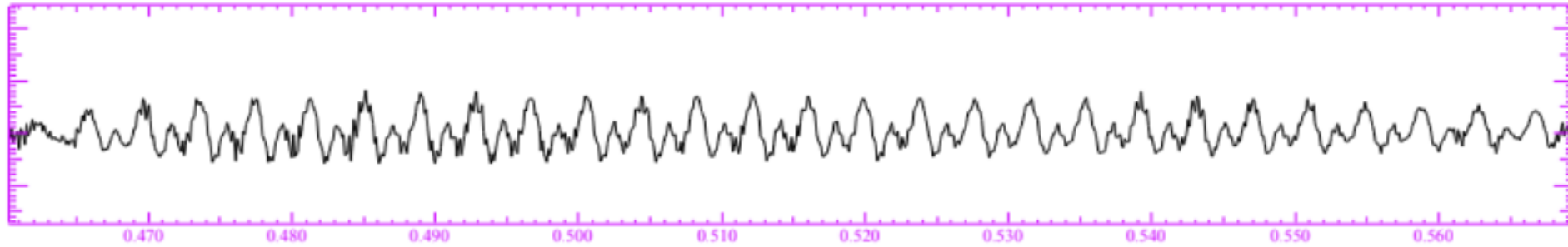
# Physical Characteristics of a Wave

## (a) Frequency

- The frequency is the **number of times a second that a wave repeats itself, or cycles.**
- Measured in **cycles per second** or **Hertz (Hz)**.
- Example: If a wave repeats 28 times in 0.11 seconds →  
Frequency =  $28 / 0.11 = 255 \text{ Hz}$
- **Higher frequency = more waves per second = higher-pitched sound.**

## (b) Amplitude

- It's the **height** of the wave (on the vertical axis).
- It shows **how powerful** the sound is, meaning how much air pressure is changing.
- **High amplitude → loud sound**  
**Low amplitude → soft sound**



**Figure 7.19** A waveform of the vowel [iy] from the utterance shown in Figure 7.20. The y-axis shows the changes in air pressure above and below normal atmospheric pressure. The x-axis shows time. Notice that the wave repeats regularly.

**Example: "She just had a baby."**

- The graph shows a **waveform** of the vowel sound [iy] (as in "she").
- The **x-axis** shows **time**, and
- The **y-axis** shows **air pressure** above or below normal atmospheric pressure.
- When the wave pattern repeats regularly → it represents a **periodic sound** (like vowels).

# Perceptual Properties of Sound

- Two important perceptual properties are related to frequency and amplitude.

## 1. Pitch

- **Correlate to: Frequency.**
- **Definition:** Higher-frequency sounds we perceive as having a higher pitch, although the relationship is not linear, since human hearing has different acuities for different frequencies.
  - High frequency → high pitch (like a whistle)
  - Low frequency → low pitch (like a drum)

## 2. Loudness

- **Correlate to: Amplitude** (specifically, the power, which is related to the *square* of the amplitude).
- **Definition:** Sounds with higher amplitudes are perceived as louder, but again, the relationship is not linear.

# How to Interpret a Waveform

- A waveform is a **graphical representation of sound**; it shows how air pressure changes over time.
  - **X-axis:** Time
  - **Y-axis:** Air pressure (above or below normal)
- So, when you see a waveform, you're looking at how loud and how frequent the sound is at each point in time.



# What Information Can We Get from a Waveform?

- Humans and machines (like speech recognizers) can study the waveform to understand speech, but visually, it gives us only *some* clues.
- Even though it's hard to decode the exact words just from the graph, we can still see:
  - Which parts are **vowels**
  - Which parts are **consonants**
  - Where there is **silence**
  - How loud or soft each sound is

# Identifying Vowels and Consonants

| Feature                | Vowels                          | Consonants                    |
|------------------------|---------------------------------|-------------------------------|
| Voicing                | Voiced (vocal folds vibrate)    | May be voiced or voiceless    |
| Length                 | Long and continuous             | Short, sometimes with silence |
| Amplitude              | High (loud)                     | Lower (soft) or irregular     |
| Appearance in Waveform | Smooth, regular repeating peaks | Irregular or broken patterns  |

## Example:

In the sentence “*She just had a baby*”:

- The vowels like [iy] (in “she”) or [ae] (in “had”) show **regular peaks** — indicating voicing.
- Consonants like [p], [t], or [k] show **silence or very low amplitude** — these are *stop consonants* (when the air is completely blocked).

# Recognizing Voicing

- **Voicing** happens when the vocal folds (in the larynx) open and close rapidly, creating vibration.  
On a waveform, voicing appears as **regular, repeating peaks** (like in vowels).
- When the vocal folds **vibrate** → **voiced sound** (e.g., [b], [d], [g], vowels)
- When there's **no vibration** → **voiceless sound** (e.g., [p], [t], [k], [s], [sh])

# Recognizing Silence and Stop Consonants

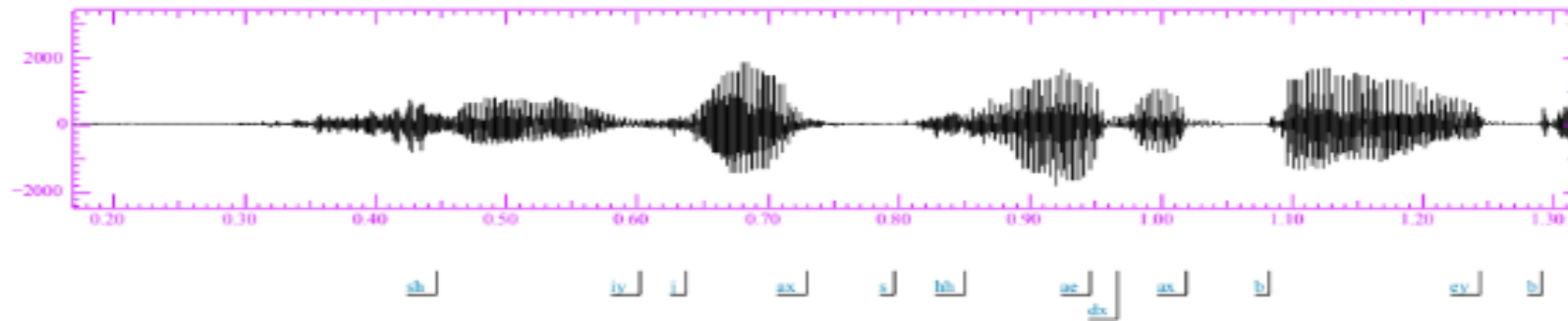
- Stop consonants involve a **complete closure** of the vocal tract — air is blocked, causing silence.
- In the example waveform (Figure 7.20):
- From **1.06–1.08 sec** (first [b]) and **1.26–1.28 sec** (second [b]) → **no peaks**, showing *stop closure (silence)*.

# Recognizing Fricatives ([sh], [f], [s])

- Fricatives are produced by **forcing air through a narrow opening**, creating noise-like, irregular patterns.

Example:

- In “*She just had a baby*”, the sound **[sh]** (from 0.33 to 0.46 sec) shows **intense, irregular peaks** — typical of fricatives.



**Figure 7.20** A waveform of the sentence “She just had a baby” from the Switchboard corpus (conversation 4325). The speaker is female, was 20 years old in 1991 which is approximately when the recording was made, and speaks the South Midlands dialect of American English. The phone labels show where each phone ends.

The waveform of “*She just had a baby*” shows:

Smooth repeating waves for vowels ([iy], [ae]) → voiced sounds.

Random noisy patterns for [sh] → fricative.

Flat silent regions for [b], [t], [k] → stop closures.

Each labeled segment shows where each **phoneme** (sound) occurs.